
Persistent Data-Order Effects Precede Directional Alignment in Language Models

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Abstract

Data order changes optimization trajectories, but endpoint divergence alone does not reveal how the corresponding trace develops or whether it is recoverable. We measure output divergence and raw parameter-space endpoint-direction alignment along public Pythia-160M/PolyPythias trajectories trained on 300 billion tokens. Three correlated order-only pairs reach symmetric KL of 0.290–0.293 nats, compared with 0.306 nats for one joint initialization-and-order pair. The central finding is a temporal dissociation: at step 16,000, the order/joint divergence ratio is already 0.932 for one pair, while its raw endpoint-direction cosine is only 0.294. These are distinct readouts, not a causal decomposition. Most parameter-block grafts then leave a predeclared language-modeling loss band, and none of the tested Fisher estimators predicts all three probe directions within 25% relative error. The comparisons use one training-distribution shard and lack an initialization-only control. Persistent order dependence is measurable. Functional recoverability and the cost of rewriting completed optimization histories remain open.

1 Introduction

Training is an ordered composition of update maps: identical initial states and data multisets can yield different completed histories. Endpoint distance leaves several questions separate. Does an output-level difference persist? When does its magnitude become substantial, and when does the raw parameter-space endpoint-difference direction align with its final direction? Can that trace be corrected, and can a proposed recovery probe be trusted?

A scalar output divergence need not track parameter orientation: later dynamics may change a displacement’s direction while preserving the scale of its effect on predictions. We examine this distinction using public Pythia and PolyPythias trajectories [1, 2], without new pretraining. Our contributions are:

1. **Temporal dissociation.** Functional divergence is near its late-training scale while the raw endpoint-difference direction remains weakly aligned: at step 16,000, the order/joint ratio is 0.932 and the cosine is 0.294 for the primary pair. Qualitative agreement across correlated pairs has pair-dependent timing.
2. **Persistent output differences.** Three correlated order-only comparisons reach 0.290–0.293 nats at the final Pythia-160M checkpoint. These are pairwise comparisons sharing one initialization, not independent replicates.

3. **Validity-gated recovery analysis.** The tested graft class and Fisher estimators fail declared validity criteria. Their failure prevents recovery claims, rather than establishing irreversibility.

2 Dynamical framework and measurement design

Transported order defects. Let x include parameters and any optimizer state, and let U_i be one update map. The adjacent-swap defect is

$$\delta_{ij}(x) = U_j(U_i(x)) - U_i(U_j(x)). \quad (1)$$

For smooth SGD updates $U_i(x) = x - \eta g_i(x)$, $\delta_{ij}(x) = \eta^2(Dg_j(x)g_i(x) - Dg_i(x)g_j(x)) + O(\eta^3)$. Thus the leading local defect is the established commutator term [5, 6]. Stateful optimizers require the augmented state. The parameter-only SGD expansion cannot be applied unchanged to Adam.

Connect two update words by adjacent swaps. For swap k , let z_k be the state after its first ordering, $z_k + \delta_k$ the state after its alternative, and A_k their common remaining update composition. Their endpoint increment is exactly

$$\Delta_k = A_k(z_k + \delta_k) - A_k(z_k), \quad x_T^{(B)} - x_T^{(A)} = \sum_k \Delta_k. \quad (2)$$

The second identity telescopes over intermediate words with a common initial state and the same update maps. Differentiability gives $\Delta_k \approx DA_k(z_k)\delta_k$: later dynamics can amplify, attenuate, or rotate local defects, which can also cancel in the sum. Checkpoint differences below are direct observations, not reconstructed per-swap attributions. The identity does not identify when Pythia’s individual order effects were generated.

A restricted correction floor. For a parameter gap Δ , fixed linear correction subspace S , and fixed quadratic metric $\Sigma \succeq 0$, define

$$r_S(\Delta) = \min_{c \in S} \|\Sigma^{1/2}(\Delta - c)\|_2. \quad (3)$$

Every correction $c \in S$ has residual norm at least $r_S(\Delta)$, directly by minimization. The minimum is attained since $\Sigma^{1/2}S$ is a finite-dimensional subspace. This elementary conditional floor is neither a nonlinear irreversibility theorem nor a computational lower bound, and does not cover state-dependent reachable spaces. Section 4 tests the metric prerequisite. No functional correction floor is measured here.

Models, metric, and controls. Write m, d_1, d_2 for the main Pythia-160M run and its two data-seed variants. They share an initialization. We analyze $P_1 = (d_1, d_2)$, $P_2 = (m, d_1)$, and $P_3 = (m, d_2)$. These are three correlated pairs from three runs, not three independent replicates. The joint arm J uses the two full-seed variants, changing initialization and order together. The endpoint is $T = 143,000$ updates (300B training tokens).

For next-token distributions on a frozen shard E , our direct readout is

$$D_E(a, b) = \frac{1}{2|E|} \sum_{x \in E} [\text{KL}(p_a(\cdot|x) \| p_b(\cdot|x)) + \text{KL}(p_b(\cdot|x) \| p_a(\cdot|x))]. \quad (4)$$

The shard contains 48 non-overlapping 2,048-token blocks from Nee1Nanda/pile-10k, concatenated with EOS separators using the Pythia tokenizer: 98,304 prediction positions. It samples the training distribution, not a held-out generalization benchmark. Models run in evaluation mode with float32 forward passes and float64 accumulation. We compare 12 checkpoints from step 0 to T , excluding step 1 after a checkpoint-integrity audit. The candidate initialization-only arm failed continuity checks and was excluded before the amended comparison. The difference $D_E(J) - D_E(P_i)$ confounds initialization and initialization-order interaction. The amended order-only criterion is weaker than the original two-arm criterion.

Positioning. Local Lie-bracket criteria predict order effects in domain and sequential learning [5, 6]. Mode-connectivity work studies instability to training noise [4]. Pythia/PolyPythias enable trajectory and seed comparisons [1, 2]. Here the focus is direct output-distribution divergence between completed histories, its timing relative to raw endpoint-direction alignment, and probe validity before recovery claims. We claim neither novelty for order sensitivity itself nor a new general transport theorem.

3 Persistent functional divergence precedes directional alignment

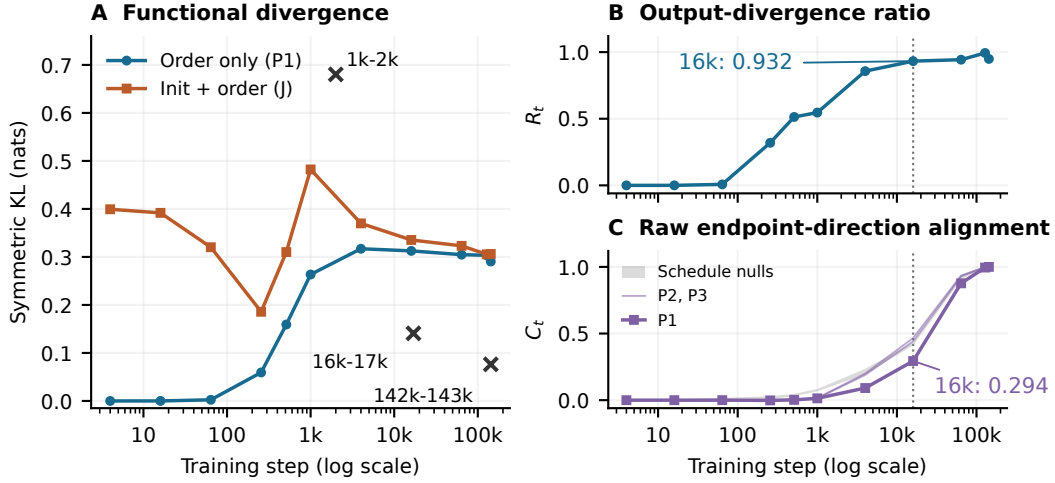


Figure 1: **Substantial functional divergence before raw endpoint-direction alignment.** Left: P_1 and joint-pair output divergence. Crosses mark drift over the labeled windows. Right: separate panels show $R_t = D_E(P_1, t)/D_E(J, t)$ and raw cosine C_t . At 16k, $R_t = 0.932$ while $C_t = 0.294$. Ratio and cosine measure different objects. Their numerical proximity has no metric interpretation. The grey band spans five schedule controls and thin lines show P_2 and P_3 . Lines connect recorded points, not fitted dynamics. Crosses are not error bars.

Late-training order effects persist. The final order-only divergences are 0.290–0.293 nats (Table 1), compared with 0.306112 for J . All exceed the late adjacent-checkpoint drift reference, 0.076275, and lie below the 160M–410M comparison, 0.460360. Their maximum/minimum ratio is 1.009, below the predeclared threshold of 2.0. The approximately 1% agreement is descriptive, not a confidence interval: each of three shared-initialization runs appears in two comparisons.

Table 1: Final-checkpoint divergence on the frozen training-distribution shard. All three order-only pairs share one initialization. Each run occurs twice. Ratios share the same single joint-pair denominator.

Pair	D_E (nats)	$D_E/D_E(J)$	$D_E/\text{drift reference}$
P_1 : data-seed1 / data-seed2	0.290461	0.9489	3.81
P_2 : main / data-seed1	0.290638	0.9494	3.81
P_3 : main / data-seed2	0.293055	0.9573	3.84
J : full-seed1 / full-seed2	0.306112	1.0000	4.01

The central dissociation: magnitude precedes raw alignment. For P_1 , let $\Delta_t = \theta_{d_1, t} - \theta_{d_2, t}$ and $C_t = \langle \Delta_t, \Delta_T \rangle / (\|\Delta_t\|_2 \|\Delta_T\|_2)$. At 16,000, $D_E(P_1) = 0.312597$ nats, already near its final scale, while $C_t = 0.294$. This statement holds in absolute KL, without the joint denominator. The ratio $R_t = D_E(P_1, t)/D_E(J, t)$ is 0.932. At 64,000, $R_t = 0.943$ and $C_t = 0.877$. The first sampled $C_t \geq 0.9$ occurs at 128,000. The ratio is not monotone: it falls from 0.993 at 128,000 to 0.949 at T .

At 16,000, P_2 and P_3 have raw cosines 0.441 and 0.466, and both first cross 0.9 at 64,000. The same qualitative dissociation occurs in all three correlated pairs, with pair-dependent timing. **Functional divergence magnitude and raw parameter-space endpoint orientation evolve on different timescales.** This is a descriptive comparison, not a test of functional-direction lock-in. Early directions are storage-quantization limited. The value $C_T = 1$ is an identity and nearby checkpoints correlate mechanically. Threshold-crossing times are grid-dependent, not precise stabilization times.

Schedule control. A low early cosine is partly expected for any trajectory. The endpoint gap includes every remaining update, and the learning rate is still high at 16,000. We therefore computed five schedule controls on the same grid. Four are single-run displacements $\cos(\theta_t - \theta_0, \theta_T - \theta_0)$ for d_1 , d_2 , and both full-seed runs. The fifth is the joint pair with its initial difference removed. At 16,000 the controls lie between 0.411 and 0.432, and all first reach 0.9 at 64,000 (grey band in Figure 1). Slow raw alignment is therefore largely a property of training under this schedule. P_1 lies 0.12 below the band, while P_2 and P_3 lie inside or above it, so we do not attribute the slow alignment to data order.

The drift reference is not a noise floor. It depends on training time and checkpoint interval: over 1,000-update windows it is 0.680225 at 1,000–2,000 and 0.140831 at 16,000–17,000. Early pair values do not establish separation from drift. The 3.8-fold comparison applies only late in training. Neither a scalar ratio nor raw cosine identifies an output-space direction or recoverability.

4 Recovery claims require validated probes

Parameter-type grafts leave the allowed loss region. Raw alignment does not measure recovery. At T , we form hybrids $\theta_h = \theta_{d_1} + \alpha M_b(\theta_{d_2} - \theta_{d_1})$, where M_b selects a parameter-type block and $\alpha \in \{0.25, 0.5, 1\}$. The frozen rule rejects token cross-entropy above the worse parent’s loss plus 0.5 nats: a threshold of 3.001560. Instrument checks passed, including parent-divergence reproduction to relative error 1.4×10^{-10} .

Seven of eight complete block swaps are invalid. The recorded loss-band tally rejects 23 of 35 evaluated graft readouts, including control and second-anchor readouts of the same hybrids. Attention QKV, carrying 87% of the squared parameter gap, is invalid at every tested nonzero strength. Only internal LayerNorm remains valid through a full swap. That hybrid is degraded and far from both parents. The frozen abort rule stops the remaining control sequence, leaving spatial recoverability inconclusive. *Parameter-space graftability cannot be assumed to measure functional recoverability:* this graft class fails at the tested strengths, without ruling out other edits. We did not run the standard control of interpolating all parameters at once [4]. These failures therefore do not establish a loss barrier between the endpoints.

Local Fisher metrics do not certify. To attempt Eq. (3) without grafts, we test $D_E(\theta, \theta + v) \approx v^\top \Sigma_E v$, with $\Sigma_E = \frac{1}{2} \hat{F}$. The predictive Fisher gives the local quadratic approximation under our half-sum KL convention, not generally the observed-label loss Hessian. Empirical and predictive Fisher are distinct [3]. Substitution needs validation.

The frozen gate requires relative prediction error at most 0.25 on all three scaled probe directions: random, the last two checkpoints’ update, and the endpoint order gap. Cross-entropy increases are 0.0050, 0.0337, and 0.0087 nats, respectively. Table 2 reports the relative errors. No family passes. The planned projection ladder and Fisher-weighted direction timing are therefore unrun.

Table 2: Local-metric certification on one endpoint pair. Each entry must be ≤ 0.25 . The predictive-Fisher diagonal uses Monte Carlo sampling. A full empirical-Fisher diagnostic also fails all three probes.

Estimator	Random	Update	Order gap	Passed
Diagonal empirical Fisher	1.11	0.77	0.47	0/3
Per-tensor empirical Gram	1.90	0.61	1.31	0/3
Diagonal predictive Fisher	0.524	0.811	0.126	1/3

The predictive diagonal has 13% error along the endpoint gap, yet underpredicts update-direction KL by about 5.3 times. *Agreement along the selected order-gap direction does not certify a local metric.* The pattern is consistent with missing cross-parameter curvature. It does not establish a unique cause. Independent builds reproduce the order-gap pass/random fail pattern, and the direct divergence readout reproduces to relative error 1.42×10^{-10} . Small loss changes alone do not certify the quadratic regime. Finite perturbations and estimator sampling remain limitations. We did not compute the exact predictive-Fisher quadratic form, which a Jacobian-vector product gives. Estimator error is therefore not separated from breakdown of the quadratic approximation. No tolerance was relaxed.

5 Discussion and limitations

Order-only output differences persist, with substantial magnitude before raw parameter-space endpoint-direction alignment. The failed validity checks separate observing a trace from measuring its recoverability. Scalar functional divergence, raw parameter direction, functional geometry, and actual correction are distinct objects. None substitutes automatically for another.

Evidence is limited to one model size, three correlated order-only pairs sharing one initialization, one joint pair, and one training-distribution shard. Initialization-only effects and interactions cannot be isolated. Independent-run statistical variation and downstream generalization are unmeasured. No universal irreversibility, successful functional transport, computational lower bound, or validity of arbitrary Fisher models follows. A next experiment should certify a metric with cross-parameter structure before projection/recovery claims, with probe criteria frozen anew. Low-rank proposals must test directions outside their represented span. Functional recoverability and cheap history correction remain open.

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